

Compatible Three-Qubit Spin Alignment Along a Diagonal Segment

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Abstract

For $q \in [1/2, 1]$, let $Q_q = \text{diag}(q, 1 - q)$. Given an arbitrary three-qubit state, form the equally weighted sum of its compatible two-body marginals, tensoring each marginal with Q_q on the complementary qubit. We prove that this operator is majorized by the corresponding aligned operator obtained by replacing every two-body marginal with $|00\rangle\langle 00|$. The proof interpolates between the maximally mixed endpoint, established by Song and Chen, and the pure endpoint. Its key point is that every Ky Fan sum of the aligned operator is affine on this diagonal segment, so Ky Fan convexity transfers the two endpoint inequalities to the whole interval. The result is limited to the equal-weight, three-qubit diagonal family; it does not settle the general compatible-marginal conjecture.

1 Setting and result

For a Hermitian matrix M , write its eigenvalues in nonincreasing order and set

$$K_k(M) := \sum_{j=1}^k \lambda_j(M).$$

For Hermitian matrices of equal trace, $M \preceq N$ means $K_k(M) \leq K_k(N)$ for every k .

Song and Chen exhibited a three-qubit counterexample to the strong spin alignment conjecture when the three two-body inputs may be chosen independently. They then formulated a compatible-marginal version and proved its three-qubit, two-body specialization when the complementary state is maximally mixed, allowing arbitrary nonnegative weights [1, Conjecture 2 and Proposition 3]. Here we extend precisely the equal-weight specialization from that endpoint to the full diagonal segment joining $I/2$ to $|0\rangle\langle 0|$.

As adjacent matrix-inequality context, Zeng et al. study sharp bounds for Schatten norms of matrix sums and give an explicit finite-dimensional counterexample [2]. Their problem is different and supplies no theorem or proof input here.

Let A, B, C be qubits, let $P_0 = |0\rangle\langle 0|$, and let ρ_{ABC} be any density matrix on $(\mathbb{C}^2)^{\otimes 3}$, with compatible marginals $\rho_{AB}, \rho_{AC}, \rho_{BC}$. Tensor factors are always placed in the order A, B, C . In particular, for operators R_{AC} and S_B , the nonadjacent embedding is fixed by

$$\langle abc|[R_{AC} \otimes S_B]_{ABC}|a'b'c'\rangle = \langle ac|R_{AC}|a'c'\rangle \langle b|S_B|b'\rangle.$$

Define

$$\begin{aligned} X_q(\rho) &:= \frac{1}{3}(\rho_{AB} \otimes Q_{q,C} + [\rho_{AC} \otimes Q_{q,B}]_{ABC} + Q_{q,A} \otimes \rho_{BC}), \\ Y_q &:= \frac{1}{3}(P_{0,A} \otimes P_{0,B} \otimes Q_{q,C} + P_{0,A} \otimes Q_{q,B} \otimes P_{0,C} + Q_{q,A} \otimes P_{0,B} \otimes P_{0,C}). \end{aligned}$$

Theorem 1. For every $q \in [1/2, 1]$ and every three-qubit density matrix ρ_{ABC} ,

$$X_q(\rho) \preceq Y_q.$$

Equivalently, $K_k(X_q(\rho)) \leq K_k(Y_q)$ for $1 \leq k \leq 7$; the traces agree when $k = 8$.

2 Endpoint interpolation

We first record the feature of the target that makes interpolation possible.

Lemma 2. In computational-basis order $000, 001, \dots, 111$,

$$Y_q = \text{diag}\left(q, \frac{1-q}{3}, \frac{1-q}{3}, 0, \frac{1-q}{3}, 0, 0, 0\right).$$

For every $q \in [1/2, 1]$ its Ky Fan sums are therefore

$$K_k(Y_q) = \begin{cases} q, & k = 1, \\ (2q+1)/3, & k = 2, \\ (q+2)/3, & k = 3, \\ 1, & 4 \leq k \leq 8. \end{cases}$$

In particular, each $K_k(Y_q)$ is affine in q throughout the interval.

Proof. The displayed diagonal follows by evaluating the three summands. The only nonzero entries are the 000 entry, equal to q , and the 001, 010, 100 entries, each equal to $(1-q)/3$. On the stated interval,

$$q - \frac{1-q}{3} = \frac{4q-1}{3} > 0, \quad \frac{1-q}{3} \geq 0,$$

so their decreasing order does not change. Summing the largest k entries gives the formula. $\square$

Proof of Theorem 1. Fix ρ_{ABC} . At $q = 1/2$, Proposition 3 of [1], with all three weights equal to $1/3$, gives the desired majorization with an identity on each complementary qubit. Multiplying both sides there by $1/2$ replaces those identities by $Q_{1/2} = I/2$ and preserves majorization. Thus

$$X_{1/2}(\rho) \preceq Y_{1/2}. \tag{1}$$

At $q = 1$, all three summands defining Y_1 coincide, and hence

$$Y_1 = |000\rangle\langle 000|.$$

Every summand defining $X_1(\rho)$ is positive semidefinite with trace one, so $X_1(\rho)$ is a density matrix. Consequently $K_k(X_1(\rho)) \leq 1 = K_k(Y_1)$ for every $k \geq 1$, and therefore

$$X_1(\rho) \preceq Y_1. \tag{2}$$

Now put $t = 2q - 1 \in [0, 1]$. Since

$$Q_q = (1-t)Q_{1/2} + tQ_1,$$

and each summand contains exactly one complementary factor, the fixed tensor embeddings give the operator identities

$$X_q(\rho) = (1-t)X_{1/2}(\rho) + tX_1(\rho), \quad Y_q = (1-t)Y_{1/2} + tY_1. \tag{3}$$

The variational formula

$$K_k(M) = \max_{\substack{P=P^*=P^2 \\ \text{rank } P=k}} \text{Tr}(PM)$$

shows that K_k is convex. Combining this convexity with (1) and (2) yields

$$\begin{aligned} K_k(X_q(\rho)) &\leq (1-t)K_k(X_{1/2}(\rho)) + tK_k(X_1(\rho)) \\ &\leq (1-t)K_k(Y_{1/2}) + tK_k(Y_1). \end{aligned}$$

By Lemma 2, the final endpoint chord is exactly $K_k(Y_q)$. This proves the required inequalities for every $1 \leq k \leq 8$. $\square$

3 Scope of the argument

Endpoint majorization and convexity alone do not generally imply majorization between interpolated operators: the target Ky Fan sum could lie strictly below its endpoint chord. Lemma 2 rules out precisely that obstruction here. The proof also fixes the state ρ_{ABC} before interpolating, so its three marginals remain jointly compatible throughout.

No conclusion is asserted for independently chosen marginals, unequal weights away from $q = 1/2$, arbitrary one-qubit states Q , higher local dimension, or additional parties. Those regimes are not consequences of the affine-chord calculation above.

Computational and AI assistance

An exact SymPy calculation with generic symbolic 4×4 marginal matrices was used to check the two affine operator identities, including the nonadjacent AC embedding, and to check the displayed target diagonal and its eight endpoint-chord identities. The calculation is supplementary; the proof above is analytic and self-contained. OpenAI Codex assisted with proof checking, symbolic verification, literature lookup, and manuscript drafting. The author reviewed the mathematical claims and presentation.

References

- [1] Zhiwei Song and Lin Chen, A counterexample to the strong spin alignment conjecture, *arXiv:2603.25410v1 [quant-ph]* (2026), <https://doi.org/10.48550/arXiv.2603.25410>.
- [2] Zijian Zeng, Houde Liu, and Kurunathan Ratnavelu, A counterexample to the Tang–Zhang Schatten norm conjecture and sharp positive results, *Preprints* (2026), <https://doi.org/10.20944/preprints202608.1067.v1>.